

Gaussian Integrals and Wick's Theorem

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November 2025

Gaussian Integration: scalar integrals

Consider the simplest Gaussian integral of scalar variable ϕ

$$\int_{-\infty}^{+\infty} d\phi e^{-\phi^2} = \sqrt{\pi}, \quad (1)$$

it can be used to compute more general ones,

$$I_1 = \int_{-\infty}^{+\infty} d\phi e^{-\frac{1}{2}a\phi^2 + \phi\psi} \quad (2)$$

$$= \int_{-\infty}^{+\infty} d\phi e^{-\frac{a}{2}\left(\phi - \frac{\psi}{a}\right)^2 + \frac{\psi^2}{2a}} \quad (3)$$

$$\stackrel{\tilde{\phi} = \sqrt{\frac{a}{2}}\left(\phi - \frac{\psi}{a}\right)}{=} e^{\frac{\psi^2}{2a}} \frac{1}{\sqrt{a/2}} \int_{-\infty}^{+\infty} d\tilde{\phi} e^{-\tilde{\phi}^2} \quad (4)$$

$$= \sqrt{\frac{2\pi}{a}} e^{\frac{1}{2a}\psi^2} \quad (5)$$

Gaussian Integration: scalar integrals

Useful identities are

$$\int_{-\infty}^{\infty} \phi^{2n} e^{-\frac{1}{2}a\phi^2} d\phi = \left(\frac{2\pi}{a}\right)^{\frac{1}{2}} \frac{1}{a^n} (2n-1)(2n-3)\cdots 5\cdot 3\cdot 1 \quad (6)$$

$$= \left(\frac{2\pi}{a}\right)^{\frac{1}{2}} \frac{1}{a^n} (2n-1)!! \quad (7)$$

Notice that we can use

$$\int_{-\infty}^{\infty} \phi^2 e^{-\frac{1}{2}a\phi^2} d\phi = -2 \frac{d}{da} \int_{-\infty}^{\infty} e^{-\frac{1}{2}a\phi^2} d\phi \quad (8)$$

$$= -2 \frac{d}{da} \left(\frac{2\pi}{a}\right)^{\frac{1}{2}} = \left(\frac{2\pi}{a}\right)^{\frac{1}{2}} \frac{1}{a} \quad (9)$$

and repeatedly apply differentiation w.r.t. a .

Gaussian Integration: N -dimensional integrals

Consider column vectors $\phi = (\phi_1, \dots, \phi_N)$ and $\psi = (\psi_1, \dots, \psi_N)$ and the $N \times N$ matrix $A = \{A_{ij}\}_{i,j=1}^N$,

$$I_N = \int_{-\infty}^{+\infty} d\phi_1 \dots d\phi_N e^{-\frac{1}{2}\phi^T A \phi + \phi^T \psi}, \quad (10)$$

where

$$-\frac{1}{2}\phi^T A \phi + \phi^T \psi = -\frac{1}{2} \sum_{ij} \phi_i A_{ij} \phi_j + \sum_i \phi_i \psi_i. \quad (11)$$

REMARK: We assume that the real matrix A is

- ① positive-definite (i.e. $\phi^T A \phi > 0$, $\forall \phi \in \mathbb{R}^N$, $\phi \neq 0$)
- ② symmetric (i.e. $A^T = A$)

Gaussian Integration: N -dimensional integrals

REMARK: If A is not symmetric, it can be symmetrized writing $A = A_{\text{sym}} + A_{\text{anti-sym}}$ with

$$A_{\text{sym}} = (A + A^T) / 2, \quad A_{\text{anti-sym}} = (A - A^T) / 2. \quad (12)$$

The anti-symmetric part generates vanishing non-diagonal terms (i.e. $\frac{1}{2}\phi^T A_{\text{anti-sym}} \phi = 0$).

EXAMPLE:

$$A = \begin{pmatrix} 1 & -1 & 5 \\ 4 & 2 & -4 \\ 1 & 2 & 1 \end{pmatrix}$$

$$A_{\text{sym}} = \begin{pmatrix} 1 & 3/2 & 3 \\ 3/2 & 2 & -1 \\ 3 & -1 & 1 \end{pmatrix}, \quad A_{\text{anti-sym}} = \begin{pmatrix} 0 & -5/2 & 2 \\ 5/2 & 0 & -3 \\ -2 & 3 & 0 \end{pmatrix}$$

Gaussian Integration: N -dimensional integrals

REMARK: A real matrix is positive definite iff its symmetric part has all (real) positive eigenvalues.

Suppose the set of eigenvalues $\{\lambda_\alpha\}_{\alpha=1}^N$ and eigenvectors $\{v_\alpha\}_{\alpha=1}^N$ of A form an orthonormal basis,

then we can express the elements of ϕ, ψ as

$$\phi_i = \sum_{\alpha} \hat{\phi}_{\alpha} v_{\alpha,i} = P \hat{\phi}, \quad \psi_i = \sum_{\alpha} \hat{\psi}_{\alpha} v_{\alpha,i} = P \hat{\psi} \quad (13)$$

with $P_{i\alpha} = v_{\alpha,i}$

$$P \hat{\phi} = \begin{pmatrix} P_{11} & \cdots & P_{1N} \\ \vdots & \ddots & \vdots \\ P_{N1} & \cdots & P_{NN} \end{pmatrix} \begin{pmatrix} \hat{\phi}_1 \\ \vdots \\ \hat{\phi}_N \end{pmatrix} = \begin{pmatrix} \sum_{\alpha} v_{\alpha,1} \hat{\phi}_{\alpha} \\ \vdots \\ \sum_{\alpha} v_{\alpha,N} \hat{\phi}_{\alpha} \end{pmatrix} \quad (14)$$

Gaussian Integration: N -dimensional integrals

Performing the change of variable $\phi \rightarrow \hat{\phi}$ and $\psi \rightarrow \hat{\psi}$, we find

$$\phi^T A \phi = (P \hat{\phi})^T A (P \hat{\phi}) = \hat{\phi}^T P^T A P \hat{\phi} = \hat{\phi}^T \Lambda \hat{\phi} = \sum_{\alpha} \lambda_{\alpha} \hat{\phi}_{\alpha}^2 \quad (15)$$

and

$$\phi^T \psi = (P \hat{\phi})^T (P \hat{\psi}) = \hat{\phi}^T P^T P \hat{\psi} = \hat{\phi}^T \hat{\psi} = \sum_{\alpha} \hat{\phi}_{\alpha} \hat{\psi}_{\alpha} \quad (16)$$

Moreover, the orthonormal (unitary) transformation has $\det P = 1$.

Gaussian Integration: N -dimensional integrals

The Gaussian integral becomes

$$I_N = \int_{-\infty}^{+\infty} d\phi_1 \dots d\phi_N e^{-\frac{1}{2}\phi^T A \phi + \phi^T \psi} \quad (17)$$

$$= \int_{-\infty}^{+\infty} \prod_{\alpha} d\hat{\phi}_{\alpha} e^{-\frac{1}{2} \sum_{\alpha} \lambda_{\alpha} \hat{\phi}_{\alpha}^2 + \sum_{\alpha} \hat{\phi}_{\alpha} \hat{\psi}_{\alpha}} \quad (18)$$

$$= \prod_{\alpha=1}^N \int_{-\infty}^{+\infty} d\hat{\phi}_{\alpha} e^{-\frac{\lambda_{\alpha}}{2} \hat{\phi}_{\alpha}^2 + \hat{\phi}_{\alpha} \hat{\psi}_{\alpha}} \quad (19)$$

$$= \prod_{\alpha=1}^N \sqrt{\frac{2\pi}{\lambda_{\alpha}}} e^{\frac{1}{2\lambda_{\alpha}} \hat{\psi}_{\alpha}^2} \quad (20)$$

$$= \sqrt{\frac{(2\pi)^N}{\det A}} e^{\frac{1}{2} \psi^T A^{-1} \psi} \quad (21)$$

Gaussian Integrals: moments and cumulants

For $N = 1$, consider the gaussian p.d.f.

$$p(\phi) = \frac{1}{Z(a, \psi)} e^{-\frac{1}{2}a\phi^2 + \phi\psi}, \quad (22)$$

where

$$Z(a, \psi) = \int_{-\infty}^{+\infty} d\phi e^{-\frac{1}{2}a\phi^2 + \phi\psi} = \sqrt{\frac{2\pi}{a}} e^{\frac{1}{2a}\psi^2} \quad (23)$$

.

The expected value of a function $f(\phi)$ is given by

$$\langle f(\phi) \rangle = \frac{\int_{-\infty}^{+\infty} d\phi f(\phi) e^{-\frac{1}{2}a\phi^2 + \phi\psi}}{\int_{-\infty}^{+\infty} d\phi e^{-\frac{1}{2}a\phi^2 + \phi\psi}}. \quad (24)$$

Gaussian Integrals: moments and cumulants

$$\langle \phi \rangle = \frac{1}{Z(a, \psi)} \frac{d}{d\psi} Z(a, \psi) = e^{-\frac{1}{2a}\psi^2} \frac{d}{d\psi} e^{\frac{1}{2a}\psi^2} \quad (25)$$

$$= \frac{\psi}{a} \quad (26)$$

$$\langle \phi^2 \rangle = \frac{1}{Z(a, \psi)} \frac{d^2}{d\psi^2} Z(a, \psi) \quad (27)$$

$$= \frac{\psi^2}{a^2} + \frac{1}{a} \quad (28)$$

A useful formula for the *moment generating function* is

$$\langle e^{-ik\phi} \rangle = e^{-ik\frac{\psi}{a} - \frac{k^2}{2a}} \quad (29)$$

Gaussian Integrals: moments and cumulants

Cumulant generating function

$$\log \langle e^{-ik\phi} \rangle \equiv \sum_{\ell=1}^{\infty} \frac{(-ik)^{\ell}}{\ell!} \langle \phi^{\ell} \rangle_c \quad (30)$$

where

$$\langle \phi \rangle_c = \langle \phi \rangle = \frac{\psi}{a} \quad (31)$$

$$\langle \phi^2 \rangle_c = \langle \phi^2 \rangle - \langle \phi \rangle^2 = 1/a \quad (32)$$

$$\langle \phi^3 \rangle_c = \langle \phi^3 \rangle - 3 \langle \phi^2 \rangle \langle \phi \rangle + 2 \langle \phi \rangle^3 \quad (33)$$

$$\langle \phi^4 \rangle_c = \langle \phi^4 \rangle - 4 \langle \phi^3 \rangle \langle \phi \rangle - 3 \langle \phi^2 \rangle^2 + 12 \langle \phi^2 \rangle \langle \phi \rangle^2 - 6 \langle \phi \rangle^4 \quad (34)$$

$\vdots$

Gaussian Integrals: moments and cumulants

$$\log \langle e^{-ik\phi} \rangle = \log \left[\sum_{\ell=0}^{\infty} \frac{(-ik)^{\ell}}{\ell!} \langle \phi^{\ell} \rangle \right] = \log \left[1 + \sum_{\ell=1}^{\infty} \frac{(-ik)^{\ell}}{\ell!} \langle \phi^{\ell} \rangle \right] \quad (35)$$

$$= \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} \left(\sum_{\ell=1}^{\infty} \frac{(-ik)^{\ell}}{\ell!} \langle \phi^{\ell} \rangle \right)^n \quad (36)$$

$$\begin{aligned} &= \left[(-ik) \langle \phi \rangle + \frac{(-ik)^2}{2!} \langle \phi^2 \rangle + \frac{(-ik)^3}{3!} \langle \phi^3 \rangle + \dots \right] \\ &\quad - \frac{1}{2} \left[(-ik) \langle \phi \rangle + \frac{(-ik)^2}{2!} \langle \phi^2 \rangle + \dots \right]^2 + \frac{1}{3} [(-ik) \langle \phi \rangle + \dots]^3 + \dots \\ &= -ik \langle \phi \rangle - \frac{k^2}{2} \langle \phi^2 \rangle + \frac{ik^3}{6} \langle \phi^3 \rangle + \frac{k^2}{2} \langle \phi \rangle^2 - \frac{ik^3}{2} \langle \phi^2 \rangle \langle \phi \rangle + \frac{i}{3} k^3 \langle \phi \rangle^3 + O(k^4) \\ &= -ik \langle \phi \rangle - \frac{k^2}{2!} [\langle \phi^2 \rangle - \langle \phi \rangle^2] + \frac{ik^3}{3!} [\langle \phi^3 \rangle - 3 \langle \phi^2 \rangle \langle \phi \rangle + 2 \langle \phi \rangle^3] + O(k^4) \\ &= \sum_{\ell=1}^{\infty} \frac{(-ik)^{\ell}}{\ell!} \langle \phi^{\ell} \rangle_c \quad (37) \end{aligned}$$

Gaussian Integrals: moments and cumulants

For $N > 1$, previous relation generalizes into

$$\left\langle e^{-i \sum_i k_i \phi_i} \right\rangle = \exp \left[-i \sum_{ij} \psi_i [A^{-1}]_{ij} k_j - \frac{1}{2} \sum_{ij} k_i [A^{-1}]_{ij} k_j \right]. \quad (38)$$

Moments and cumulants are obtained by differentiation, e.g.

$$\langle \phi_i \rangle_c = i \frac{\partial}{\partial k_i} \left\langle e^{-i \sum_j k_j \phi_j} \right\rangle \Big|_{k_j=0, \forall j} \quad (39)$$

$$= \sum_j [A^{-1}]_{ij} \psi_j, \quad (40)$$

$$\langle \phi_i \phi_j \rangle_c = - \frac{\partial}{\partial k_j} \frac{\partial}{\partial k_i} \left\langle e^{-i \sum_j k_j \phi_j} \right\rangle \Big|_{k_j=0, \forall j} - \langle \phi_i \rangle \langle \phi_j \rangle \quad (41)$$

$$= [A^{-1}]_{ij}. \quad (42)$$

Gaussian Integrals: moments and cumulants

$$\langle e^{-i \sum_i k_i \phi_i} \rangle = \frac{\int_{-\infty}^{+\infty} d\phi_1 \dots d\phi_N e^{-i\phi^T k} e^{-\frac{1}{2}\phi^T A \phi + \phi^T \psi}}{\sqrt{\frac{(2\pi)^N}{\det A}} e^{\frac{1}{2}\psi^T A^{-1} \psi}} \quad (43)$$

$$= \frac{\int_{-\infty}^{+\infty} d\phi_1 \dots d\phi_N e^{-\frac{1}{2}\phi^T A \phi + \phi^T (\psi - ik)}}{\sqrt{\frac{(2\pi)^N}{\det A}} e^{\frac{1}{2}\psi^T A^{-1} \psi}} \quad (44)$$

$$= \frac{\sqrt{\frac{(2\pi)^N}{\det A}} e^{\frac{1}{2}(\psi - ik)^T A^{-1} (\psi - ik)}}{\sqrt{\frac{(2\pi)^N}{\det A}} e^{\frac{1}{2}\psi^T A^{-1} \psi}} \quad (45)$$

$$= \frac{e^{\frac{1}{2}(\psi - ik)^T A^{-1} (\psi - ik)}}{e^{\frac{1}{2}\psi^T A^{-1} \psi}} \quad (46)$$

$$= \exp \left[-i \sum_{ij} \psi_i A_{ij}^{-1} k_j - \frac{1}{2} \sum_{ij} k_i A_{ij}^{-1} k_j \right] \quad (47)$$

Functional Gaussian Integrals

In the appropriate limit $N \rightarrow \infty$, we obtain functional integrals

$$\mathcal{I} = \int D\phi \, e^{-\frac{1}{2} \int d\vec{x} d\vec{x}' \phi(\vec{x}) K(\vec{x}, \vec{x}') \phi(\vec{x}') + \int d\vec{x} \psi(\vec{x}) \phi(\vec{x})} \quad (48)$$

$$= \mathcal{N}(K) \exp \left[\frac{1}{2} \int d\vec{x} d\vec{x}' \psi(\vec{x}) K^{-1}(\vec{x}, \vec{x}') \psi(\vec{x}') \right] \quad (49)$$

in which the inverse of the linear operator $K(\vec{x}, \vec{y})$ is defined by

$$\int d\vec{x}'' K^{-1}(\vec{x}, \vec{x}'') K(\vec{x}'', \vec{x}') = \delta(\vec{x} - \vec{x}'). \quad (50)$$

The prefactor is given by

$$\mathcal{N}(K) = \lim_{N \rightarrow \infty} (2\pi)^{N/2} (\det K)^{-1/2}. \quad (51)$$

but it does not appear in measurable quantities.

Functional Gaussian Integrals

Multi-point correlation functions by functional differentiation

$$\begin{aligned}\langle \phi(\vec{x}) \rangle_c &= \frac{\int D\phi \phi(\vec{x}) e^{-\frac{1}{2} \int d\vec{x} d\vec{x}' \phi(\vec{x}) K(\vec{x}, \vec{x}') \phi(\vec{x}') + \int d\vec{x} \psi(\vec{x}) \phi(\vec{x})}}{\int D\phi e^{-\frac{1}{2} \int d\vec{x} d\vec{x}' \phi(\vec{x}) K(\vec{x}, \vec{x}') \phi(\vec{x}') + \int d\vec{x} \psi(\vec{x}) \phi(\vec{x})}} \\&= \frac{\frac{\delta}{\delta \psi(\vec{x})} e^{\frac{1}{2} \int d\vec{x} d\vec{x}' \psi(\vec{x}) K^{-1}(\vec{x}, \vec{x}') \psi(\vec{x}')}}{e^{\frac{1}{2} \int d\vec{x} d\vec{x}' \psi(\vec{x}) K^{-1}(\vec{x}, \vec{x}') \psi(\vec{x}')}} \\&= \frac{1}{\mathcal{I}} \frac{\delta \mathcal{I}[\psi]}{\delta \psi(\vec{x})} = \int d\vec{x}' K^{-1}(\vec{x}, \vec{x}') \psi(\vec{x}')\end{aligned}$$

$$\langle \phi(\vec{x}) \phi(\vec{x}') \rangle_c = \langle \phi(\vec{x}) \phi(\vec{x}') \rangle - \langle \phi(\vec{x}) \rangle \langle \phi(\vec{x}') \rangle \quad (52)$$

$$= \frac{1}{\mathcal{I}} \frac{\delta^2 \mathcal{I}[\psi]}{\delta \psi(\vec{x}) \delta \psi(\vec{x}')} - \left(\frac{1}{\mathcal{I}} \frac{\delta \mathcal{I}[\psi]}{\delta \psi(\vec{x})} \right) \left(\frac{1}{\mathcal{I}} \frac{\delta \mathcal{I}[\psi]}{\delta \psi(\vec{x}')} \right) \quad (53)$$

$$= \frac{\delta}{\delta \psi(\vec{x}')} \left[\frac{1}{\mathcal{I}} \frac{\delta \mathcal{I}[\psi]}{\delta \psi(\vec{x})} \right] = K^{-1}(\vec{x}, \vec{x}') \quad (54)$$

Wick's Theorem

Setting $\psi_i = 0 \forall i$ and calling $a_i = -ik_i$

$$\left\langle e^{\sum_i a_i \phi_i} \right\rangle = e^{\frac{1}{2} \sum_{ij} a_i \langle \phi_i \phi_j \rangle a_j} \quad (55)$$

The l.h.s. is

$$\begin{aligned} \left\langle e^{\sum_i a_i \phi_i} \right\rangle &= 1 + \sum_i a_i \langle \phi_i \rangle + \frac{1}{2} \sum_{ij} a_i a_j \langle \phi_i \phi_j \rangle + \frac{1}{6} \sum_{ijk} a_i a_j a_k \langle \phi_i \phi_j \phi_k \rangle \\ &\quad + \frac{1}{24} \sum_{ijkl} a_i a_j a_k a_l \langle \phi_i \phi_j \phi_k \phi_l \rangle + \dots \end{aligned} \quad (56)$$

The r.h.s. instead

$$e^{\frac{1}{2} \sum_{ij} a_i \langle \phi_i \phi_j \rangle a_j} = 1 + \frac{1}{2} \sum_{ij} a_i a_j \langle \phi_i \phi_j \rangle + \frac{1}{8} \left(\sum_{ij} a_i a_j \langle \phi_i \phi_j \rangle \right)^2 + \dots \quad (57)$$

Wick's Theorem

Comparing term by term, we have the relation

$$\left\langle \prod_{i=1}^k \phi_i \right\rangle = \begin{cases} 0 & k = 2n + 1 \\ \sum_P \langle \phi_{p_1} \phi_{p_2} \rangle \langle \phi_{p_3} \phi_{p_4} \rangle \cdots \langle \phi_{p_{2n-1}} \phi_{p_{2n}} \rangle & k = 2n, \end{cases}$$

where P indicates all possible permutations of indexes (sum over all *pairwise contractions*, there are $(2n - 1)!!$ terms).

For gaussian fields, the multi-point correlation functions reduce to

$$\langle \phi(x_1) \phi(x_2) \cdots \phi(x_{2n}) \rangle = \sum_P \langle \phi(x_{p_1}) \phi(x_{p_2}) \rangle \langle \phi(x_{p_3}) \phi(x_{p_4}) \rangle \cdots \langle \phi(x_{p_{2n-1}}) \phi(x_{p_{2n}}) \rangle \quad (58)$$

$$= \sum_P G(x_{p_1}, x_{p_2}) G(x_{p_3}, x_{p_4}) \cdots G(x_{p_{2n-1}}, x_{p_{2n}}), \quad (59)$$

REMARK: This property is a special case of more general *Wick's Theorem* in Quantum Field Theory and operator algebra.

Application: another derivation of $\mathcal{H}_{\text{eff}}$

Let us consider a microscopic Ising model, to which we apply the gaussian integral identity

$$\int_{-\infty}^{+\infty} \prod_i d\psi_i \exp \left[-\frac{1}{4\beta} \sum_{ij} \psi_i J_{ij}^{-1} \psi_j + \sum_i \psi_i s_i \right] = \mathcal{N}^{-1} e^{\beta \sum_{ij} J_{ij} s_i s_j}, \quad (60)$$

where $\mathcal{N}^{-1} = \sqrt{\frac{(4\pi\beta)^N}{\det(J^{-1})}}$. Then the partition function becomes

$$Z = \sum_{\vec{s}} e^{\beta \sum_{ij} J_{ij} s_i s_j + \beta \sum_i h_i s_i} \quad (61)$$

$$= \mathcal{N} \sum_{\vec{s}} \int_{-\infty}^{+\infty} \prod_{i=1}^N d\psi_i e^{-\frac{1}{4\beta} \sum_{ij} \psi_i J_{ij}^{-1} \psi_j + \sum_i (\psi_i + \beta h_i) s_i} \quad (62)$$

Application: another derivation of $\mathcal{H}_{\text{eff}}$

Introducing $\tilde{\psi}_i = \psi_i + \beta h_i$,

$$\begin{aligned} Z &= \mathcal{N} \int_{-\infty}^{+\infty} \prod_{i=1}^N d\tilde{\psi}_i \exp \left[-\frac{1}{4\beta} \sum_{ij} (\tilde{\psi}_i - \beta h_i) J_{ij}^{-1} (\tilde{\psi}_j - \beta h_j) \right] \\ &\quad \times \sum_{\vec{s}} e^{\sum_i \tilde{\psi}_i s_i} \end{aligned} \quad (63)$$

$$= \mathcal{N} \int_{-\infty}^{+\infty} \prod_{i=1}^N d\tilde{\psi}_i e^{-\frac{1}{4\beta} \sum_{ij} (\tilde{\psi}_i - \beta h_i) J_{ij}^{-1} (\tilde{\psi}_j - \beta h_j) + \sum_i \log(2 \cosh \tilde{\psi}_i)} \quad (64)$$

With the linear transformation $\phi_i = \frac{1}{2\beta} \sum_j J_{ij}^{-1} \tilde{\psi}_j$ (and $\tilde{\psi}_i = 2\beta \sum_j J_{ij} \phi_j$),

$$\sum_{ij} (\tilde{\psi}_i - \beta h_i) J_{ij}^{-1} (\tilde{\psi}_j - \beta h_j) = \sum_{ij} \tilde{\psi}_i J_{ij}^{-1} \tilde{\psi}_j - 2\beta \sum_{ij} h_i J_{ij}^{-1} \tilde{\psi}_j + \beta^2 \sum_{ij} h_i J_{ij}^{-1} h_j \quad (65)$$

$$= 4\beta^2 \sum_{ij} \phi_i J_{ij} \phi_j - 4\beta^2 \sum_i h_i \phi_i + \beta^2 \sum_{ij} h_i J_{ij}^{-1} h_j \quad (66)$$

Application: another derivation of $\mathcal{H}_{\text{eff}}$

The partition function becomes

$$\begin{aligned} Z &= \mathcal{N} 2^N e^{-\frac{\beta}{4} \sum_{ij} h_i J_{ij}^{-1} h_j} \\ &\quad \times \int_{-\infty}^{+\infty} \prod_{i=1}^N d\phi_i e^{-\beta \sum_{ij} \phi_i J_{ij} \phi_j + \beta \sum_i h_i \phi_i + \sum_i \log \cosh(2\beta \sum_k J_{ik} \phi_k)} \\ &\propto \int_{-\infty}^{+\infty} \prod_{i=1}^N d\phi_i e^{-\beta \sum_{ij} \phi_i J_{ij} \phi_j + \beta \sum_i h_i \phi_i + \sum_i \log \cosh(2\beta \sum_k J_{ik} \phi_k)}, \quad (67) \end{aligned}$$

in which we can identify an energetic term and an entropic one.

We can proceed either by performing coarse-graining or moving to the Fourier space.

Application: another derivation of $\mathcal{H}_{\text{eff}}$

EXERCISE: Using

$$\phi_i = \phi(\vec{x}_i) = \frac{1}{V} \sum_{\vec{k}} \hat{\phi}(\vec{k}) e^{i\vec{k} \cdot \vec{x}_i} \quad (68)$$

$$h_i = h(\vec{x}_i) = \frac{1}{V} \sum_{\vec{k}} \hat{h}(\vec{k}) e^{i\vec{k} \cdot \vec{x}_i} \quad (69)$$

$$J_{ij} = J(\vec{x}_i - \vec{x}_j) = \frac{1}{V} \sum_{\vec{k}} \hat{J}(\vec{k}) e^{i\vec{k} \cdot (\vec{x}_i - \vec{x}_j)}, \quad (70)$$

show that

$$\sum_{ij} \phi_i J_{ij} \phi_j = \frac{1}{V} \sum_{\vec{k}} \hat{J}(\vec{k}) \hat{\phi}(\vec{k}) \hat{\phi}(-\vec{k}) = \frac{1}{V} \sum_{\vec{k}} \hat{J}(\vec{k}) |\hat{\phi}(\vec{k})|^2 \quad (71)$$

and

$$\sum_i h_i \phi_i = \frac{1}{V} \sum_{\vec{k}} \hat{\phi}(\vec{k}) \hat{h}(-\vec{k}). \quad (72)$$

Application: another derivation of $\mathcal{H}_{\text{eff}}$

EXERCISE: Using the series expansion $\log \cosh x = \frac{1}{2}x^2 - \frac{1}{12}x^4 + \dots$, the “entropic term” becomes

$$\sum_i \log \cosh \left(2\beta \sum_j J_{ij} \phi_j \right) \approx 2\beta^2 \sum_i \left(\sum_j J_{ij} \phi_j \right)^2 - \frac{4\beta^4}{3} \sum_i \left(\sum_j J_{ij} \phi_j \right)^4 \quad (73)$$

Show that

$$\sum_i \left(\sum_j J_{ij} \phi_j \right)^2 = \frac{1}{V} \sum_{\vec{k}} |\hat{J}(\vec{k})|^2 |\hat{\phi}(\vec{k})|^2 \quad (74)$$

$$\sum_i \left(\sum_j J_{ij} \phi_j \right)^4 = \frac{1}{V^4} \sum_{\vec{k}_1, \vec{k}_2, \vec{k}_3, \vec{k}_4} \delta \left(\sum_{i=1}^4 \vec{k}_i \right) \prod_{i=1}^4 \hat{J}(\vec{k}_i) \hat{\phi}(\vec{k}_i) \quad (75)$$

Application: another derivation of $\mathcal{H}_{\text{eff}}$

Collecting everything

$$Z \propto \int \prod_{\vec{k}} d\hat{\phi}(\vec{k}) d\hat{\phi}^*(\vec{k}) \exp \left[-\frac{\beta}{V} \sum_{\vec{k}} \left(\hat{J}(\vec{k}) - 2\beta |\hat{J}(\vec{k})|^2 \right) |\hat{\phi}(\vec{k})|^2 \right. \\ \left. - \frac{4\beta^4}{3V^4} \sum_{\vec{k}_1, \vec{k}_2, \vec{k}_3, \vec{k}_4} \delta \left(\sum_{i=1}^4 \vec{k}_i \right) \prod_{i=1}^4 \hat{J}(\vec{k}_i) \hat{\phi}(\vec{k}_i) + \frac{\beta}{V} \sum_{\vec{k}} \hat{\phi}(\vec{k}) \hat{h}(-\vec{k}) \right] \quad (76)$$

If the field is smooth at sufficiently large scale, we can approximate

$$\hat{J}(\vec{k}) = \int d\vec{r} J(\vec{r}) e^{-i\vec{k} \cdot \vec{r}} \quad (77)$$

$$= \int d\vec{r} J(\vec{r}) - \frac{1}{2} k^2 \int d\vec{r} r^2 J(\vec{r}) + O(k^4) \quad (78)$$

$$= J_0 - J_2 k^2 + O(k^4) \quad (79)$$

Application: another derivation of $\mathcal{H}_{\text{eff}}$

First term is given by

$$\frac{\beta}{V} \sum_{\vec{k}} \left(\hat{J}(\vec{k}) - 2\beta |\hat{J}(\vec{k})|^2 \right) |\hat{\phi}(\vec{k})|^2 \quad (80)$$

$$= \frac{\beta}{V} \sum_{\vec{k}} \left[J_0 - J_2 k^2 - 2\beta (J_0 - J_2 k^2)^2 \right] |\hat{\phi}(\vec{k})|^2 \quad (81)$$

$$= \frac{\beta}{V} \sum_{\vec{k}} J_0 \left[1 - \frac{J_2}{J_0} k^2 - 2\beta (J_0 - 2J_2 k^2) \right] |\hat{\phi}(\vec{k})|^2 \quad (82)$$

$$= \frac{\beta}{V} \sum_{\vec{k}} J_0 \left[1 - 2\beta J_0 + (4\beta J_0 - 1) \frac{J_2}{J_0} k^2 \right] |\hat{\phi}(\vec{k})|^2 \quad (83)$$

Application: another derivation of $\mathcal{H}_{\text{eff}}$

Second term is given by

$$-\frac{4\beta^4}{3V^4} \sum_{\vec{k}_1, \vec{k}_2, \vec{k}_3, \vec{k}_4} \delta \left(\sum_{i=1}^4 \vec{k}_i \right) \prod_{i=1}^4 \hat{J}(\vec{k}_i) \hat{\phi}(\vec{k}_i) \quad (84)$$

$$= -\frac{4\beta^4}{3V^4} \sum_{\vec{k}_1, \vec{k}_2, \vec{k}_3, \vec{k}_4} \delta \left(\sum_{i=1}^4 \vec{k}_i \right) \prod_{i=1}^4 (J_0 - J_2 k_i^2) \hat{\phi}(\vec{k}_i) \quad (85)$$

$$\approx -\frac{4\beta^4 J_0^4}{3V^4} \sum_{\vec{k}_1, \vec{k}_2, \vec{k}_3} \hat{\phi}(\vec{k}_1) \hat{\phi}(\vec{k}_2) \hat{\phi}(\vec{k}_3) \hat{\phi}(-\vec{k}_1 - \vec{k}_2 - \vec{k}_3) \quad (86)$$

Application: another derivation of $\mathcal{H}_{\text{eff}}$

The resulting effective Hamiltonian is of the type

$$\begin{aligned} \beta \mathcal{H}_{\text{eff}} \approx & \frac{1}{V} \sum_{\vec{k}} \left[\frac{\hat{r}_0 \hat{t}}{2} + \frac{\hat{c}_0}{2} k^2 \right] |\hat{\phi}(\vec{k})|^2 - \frac{\beta}{V} \sum_{\vec{k}} \hat{\phi}(\vec{k}) \hat{h}(-\vec{k}) \\ & + \frac{\hat{u}_0}{4} \sum_{\vec{k}_1, \vec{k}_2, \vec{k}_3} \hat{\phi}(\vec{k}_1) \hat{\phi}(\vec{k}_2) \hat{\phi}(\vec{k}_3) \hat{\phi}(-\vec{k}_1 - \vec{k}_2 - \vec{k}_3). \end{aligned} \quad (87)$$

Inverting the Fourier transform,

$$\beta \mathcal{H}_{\text{eff}}[\phi] = \int d\vec{x} \left\{ \frac{\hat{c}_0}{2} (\nabla \phi)^2 + \frac{\hat{r}_0 \hat{t}}{2} \phi(\vec{x})^2 + \frac{\hat{u}_0}{4} \phi(\vec{x})^4 - \beta \phi(\vec{x}) h(\vec{x}) \right\}. \quad (88)$$